Financial Inclusion of the Informal Sector of Marginalized Counties in Kenya

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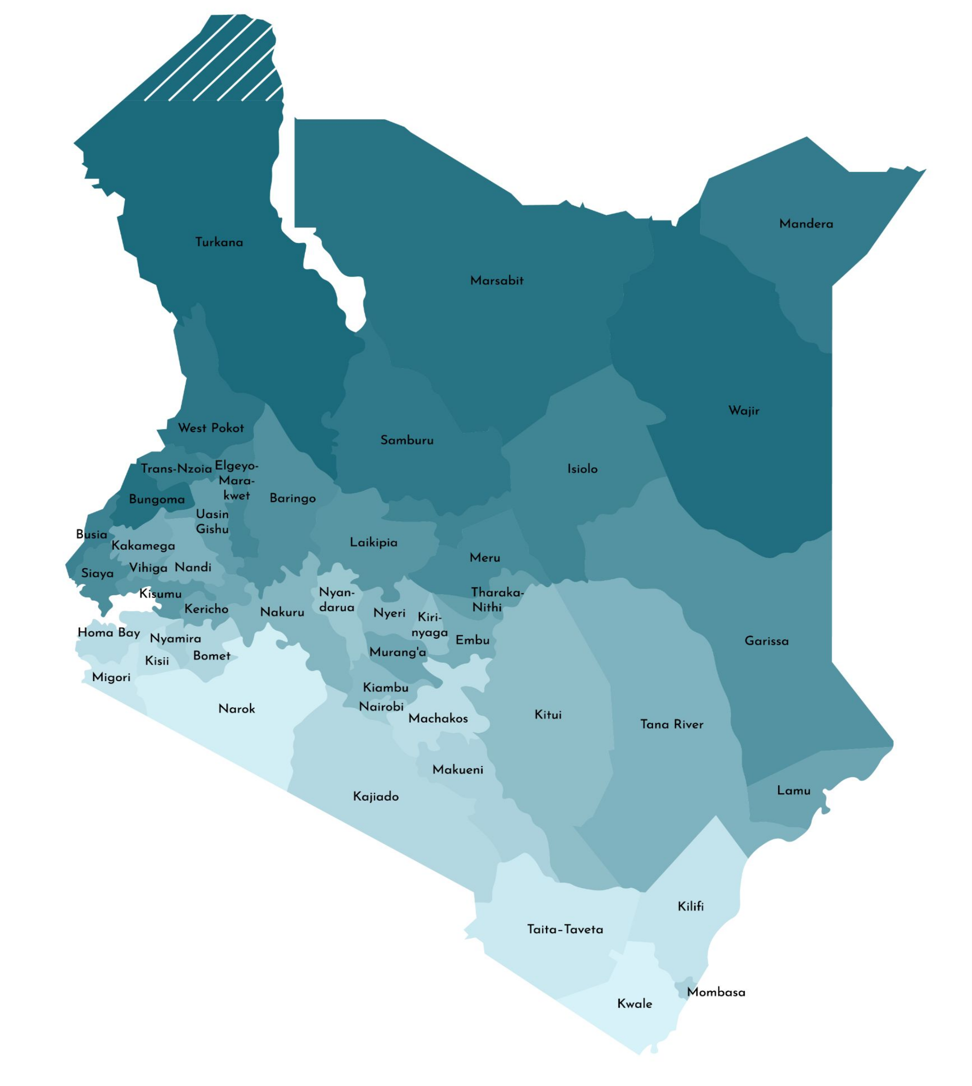
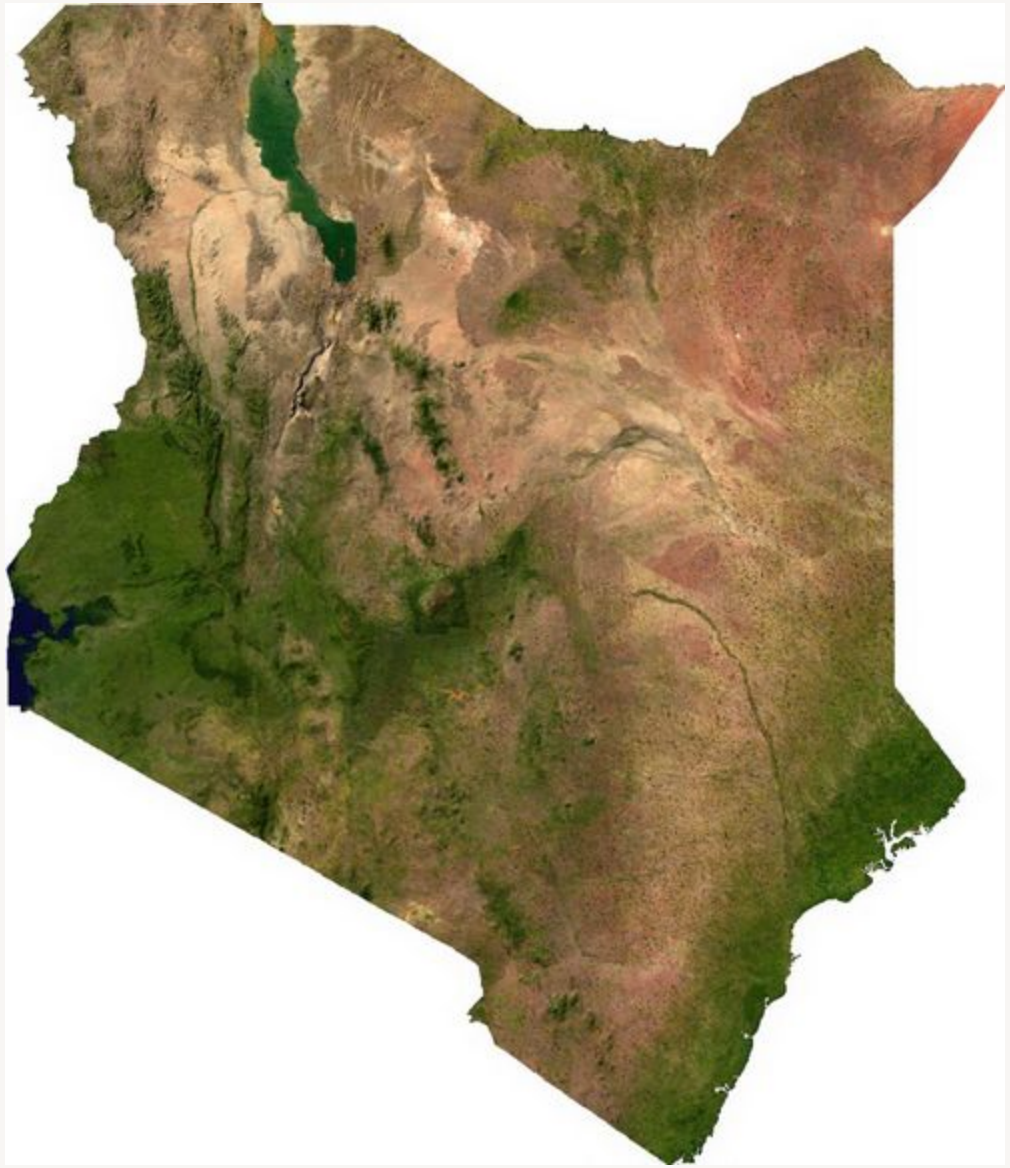


Introduction

- The 2024 FinAccess Household Survey shows Kenya's financial inclusion continuing to edge upward, with **84.8% of adults now formally served** and complete exclusion down to **9.9%**.
- Mobile money remains the backbone of access, driving a sharp rise in **daily digital transactions (over 50% of adults)** and narrowing gender and rural–urban gaps.
- Banks, SACCOs, and government initiatives such as the Hustler Fund have strengthened formal usage, yet growth is slowing, signalling a possible plateau.
- Key gaps persist among **rural youth, persons with disabilities, and those lacking IDs or phones**, while only **one in five Kenyans is financially healthy** enough to withstand shocks or invest for the future.
- The survey calls for targeted policies to deepen usage, enhance product quality, and expand sustainable and green finance so that inclusion translates into lasting financial wellbeing.



The Challenge

Aim:

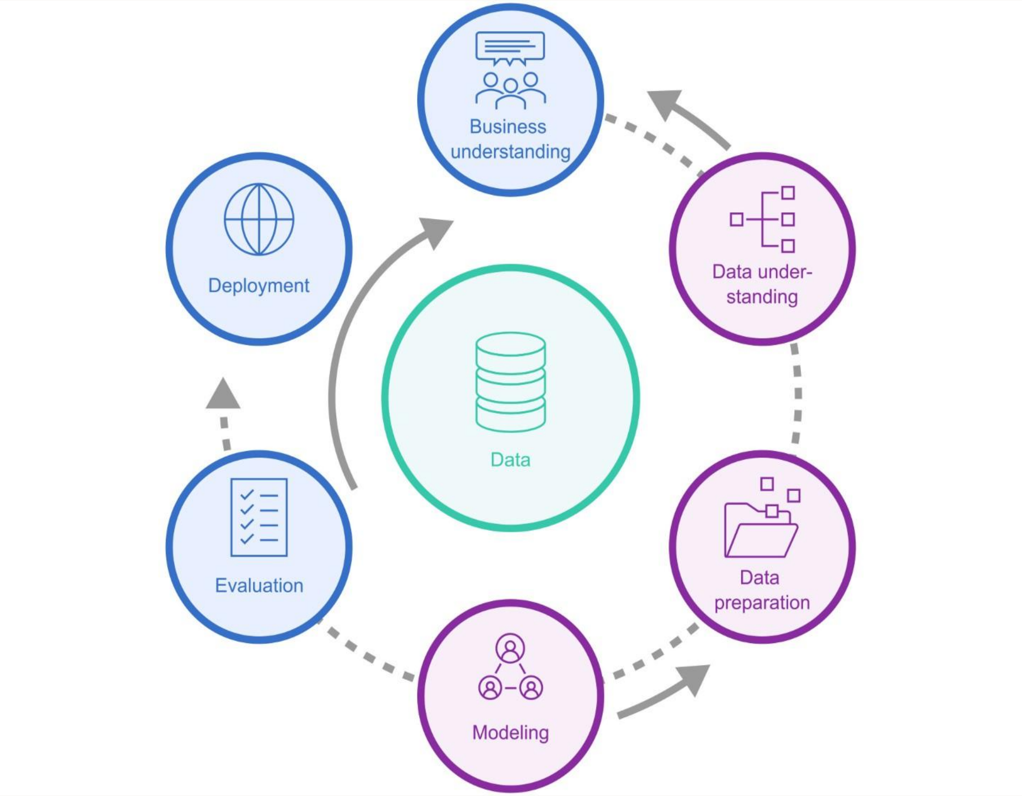
Investigate the depth of financial inclusion within the informal sector of marginalized counties.

Key Questions:

- What drives or limits formal financial inclusion for small informal businesses?
- How do mobile money and banking services interact with local realities?
- Which interventions—policy or fintech—can close the gap?

CRISP-DM Methodology

1. **Business Understanding** Goal: Identify drivers & inhibitors of financial inclusion for informal sector.
2. **Data Understanding** FinAccess 2024 micro-data + 320 structured questionnaires + qualitative interviews.
3. **Data Preparation** Cleaning, missing-value treatment, feature engineering (demographics, transaction patterns).
4. **Modeling** Probit model for inclusion likelihood.Clustering for segment discovery.
5. **Evaluation** Accuracy, interpretability, policy relevance.
6. **Deployment** Final outcome: **Interactive Dashboard** for stakeholders.



About the Dataset

Primary Data:

- Structured questionnaires of **320 respondents** (Jua Kali traders)
- semi-structured interviews with entrepreneurs, local financial service providers, and officials.

Secondary Data:

- **FinAccess 2024 dataset**
- (KNBS portal) for national and county-level benchmarks.

Study Design & Sampling

- **Counties:** Turkana, Mandera, Wajir, Marsabit, Garissa, Samburu, Tana River, Isiolo, West Pokot.
- **Method:** Purposive selection (<70% inclusion) → **stratified random sampling** of 320 respondents.
- **Criteria:** Annual turnover KSh 0.5–2 M, sole proprietors or ≤2 employees, age ≥ 18, no formal salary.

Data Preparation

Data was integrated from different sources i.e financial services, demographics data ,mobile services and barrier data.

Data cleaning was done to remove missing values and outliers

Interview_ID column was dropped because it doesn’t contribute to the prediction in our financial model.

```
Missing values per column:
County                                          0
Residence                                       0
Gender                                          0
PWD                                             0
Age                                             0
ID_card                                         0
livelihoodcat                                   0
Education                                       0
Quintiles                                       0
Marital                                         0
Cash                                            0
Mobile money (Send money) (M-Pesa, Airtel Money 0
Mobile money business wallet                    0
Bank/SACCO/MFI paybill                          0
Merchant/business paybill/till number           0
Bank cheque                                     0
Bank transfer (e.g. EFT, SWIFT, Pesalink)       0
Credit cards / debit cards                      0
No cash                                         0
Digital /community currency e.g bitcoin         0
Other Money Transfer methods                    0
dtype: int64

Data types of columns:
...
No cash                                         int64
Digital /community currency e.g bitcoin         int64
Other Money Transfer methods                    int64
dtype: object
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Data Quality Report

Initial dataset size: 20871 rows, 3816 columns.

Cleaned dataset size: 3210 rows, 22 columns.

Variables: Selected variables capture key demographic, socioeconomic, and financial dimensions influencing financial inclusion in marginalized counties. They encompass personal identifiers(age, gender, residence, disability), economic factors (education, livelihood, wealth), and financial access indicators(banking, savings, credit, microfinance).

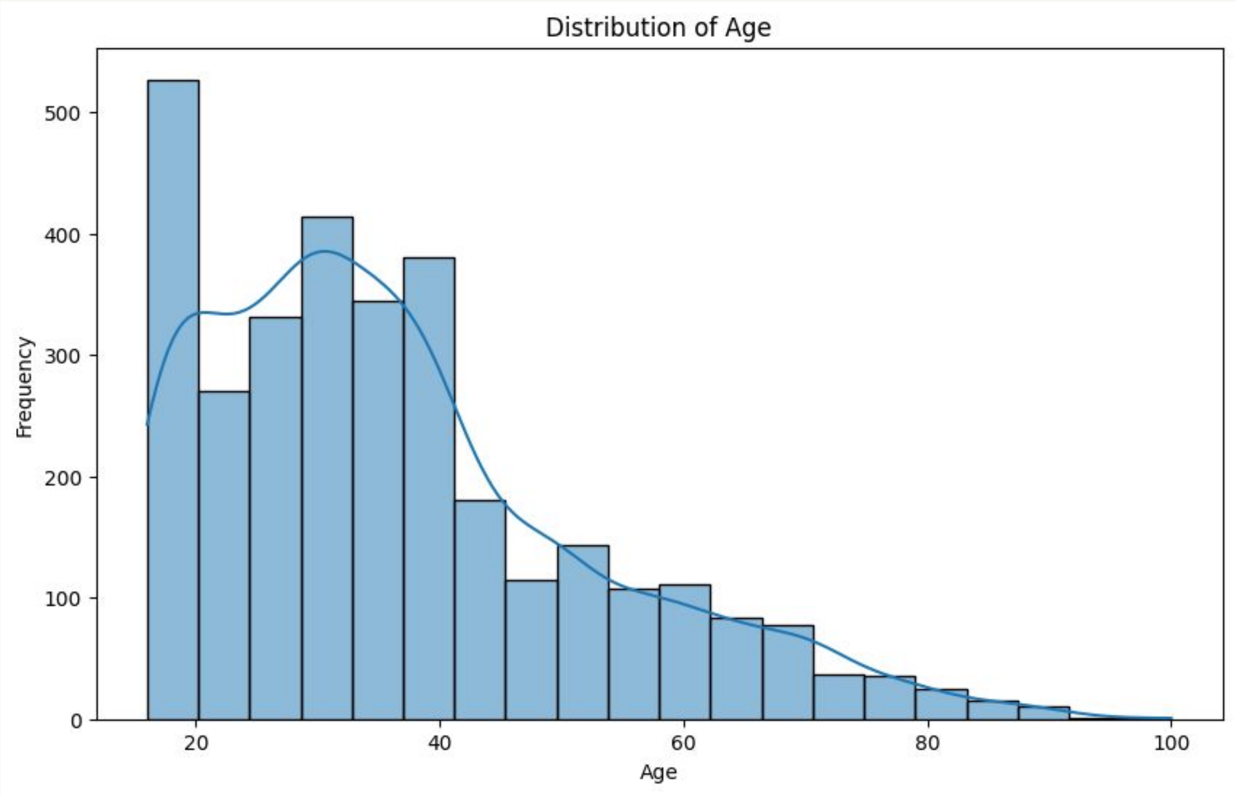
Completeness: Minimal missing data (<5%) across all variables.

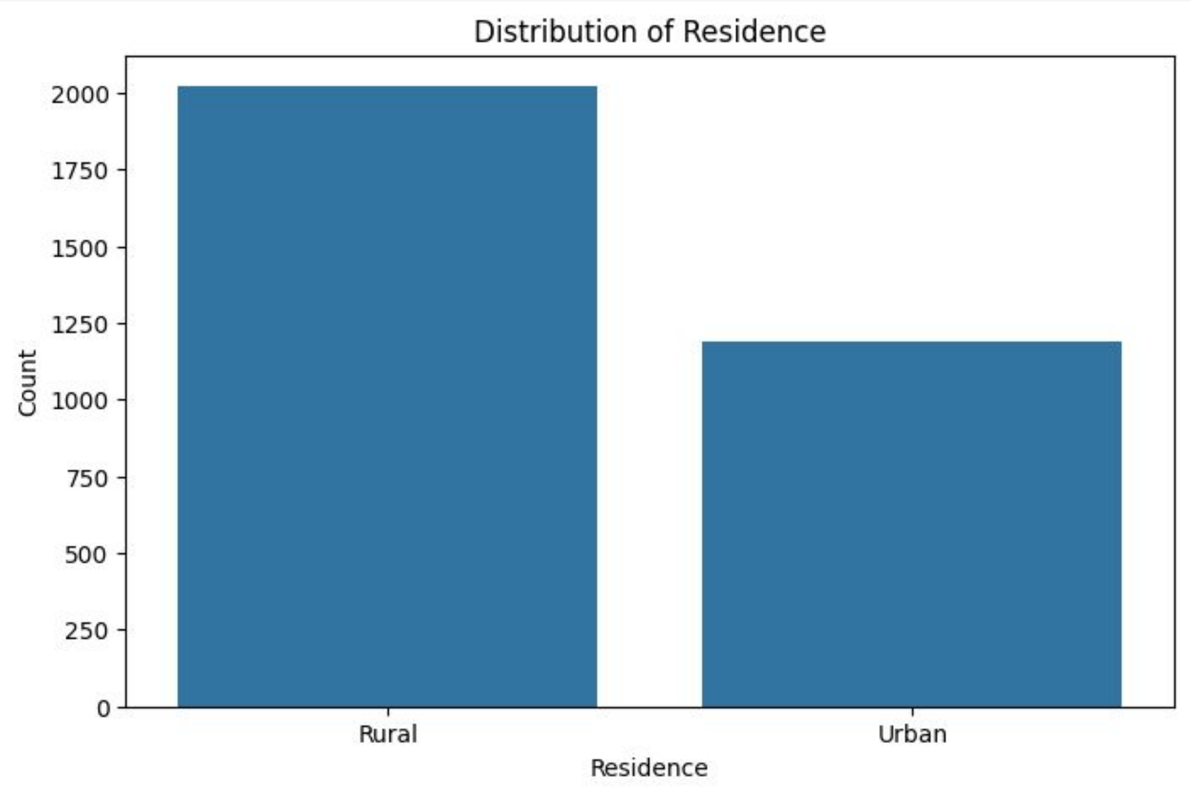
Uniqueness: All respondents uniquely identified by ‘interview_id’

Data types: Amix of categorical (Gender, county, residence, education, livelihood, financial access indicators) and numerical (age, wealth, quintiles)

Data integrity: No out of range or invalid entities for key variables.

Data Exploration

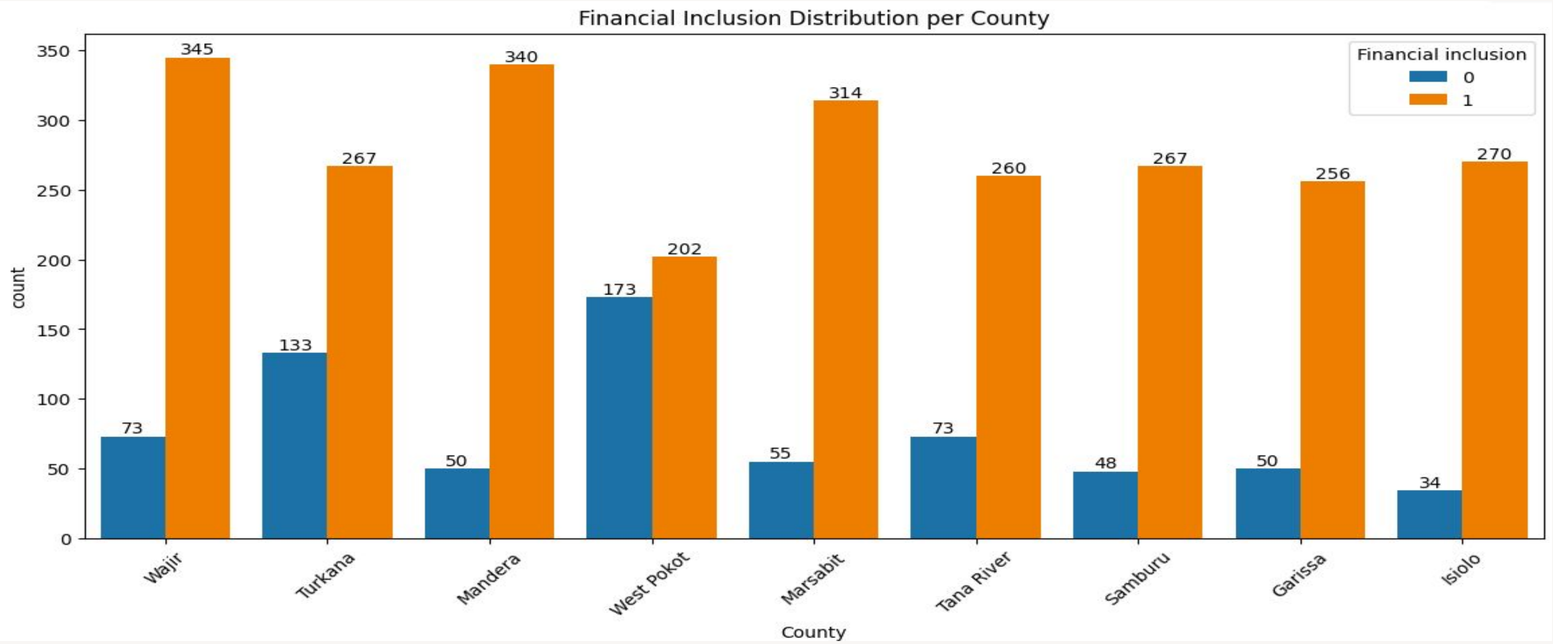


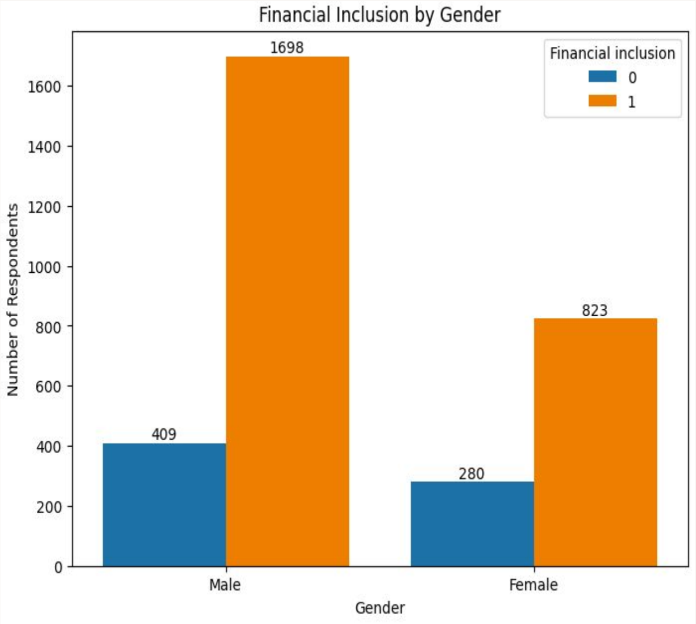


. The Age Distribution is skewed to the right as majority of the residence are between 20 yrs to 45 yrs old.

. Most of the residence live in Rural area compare to urban area.

Demographic Patterns

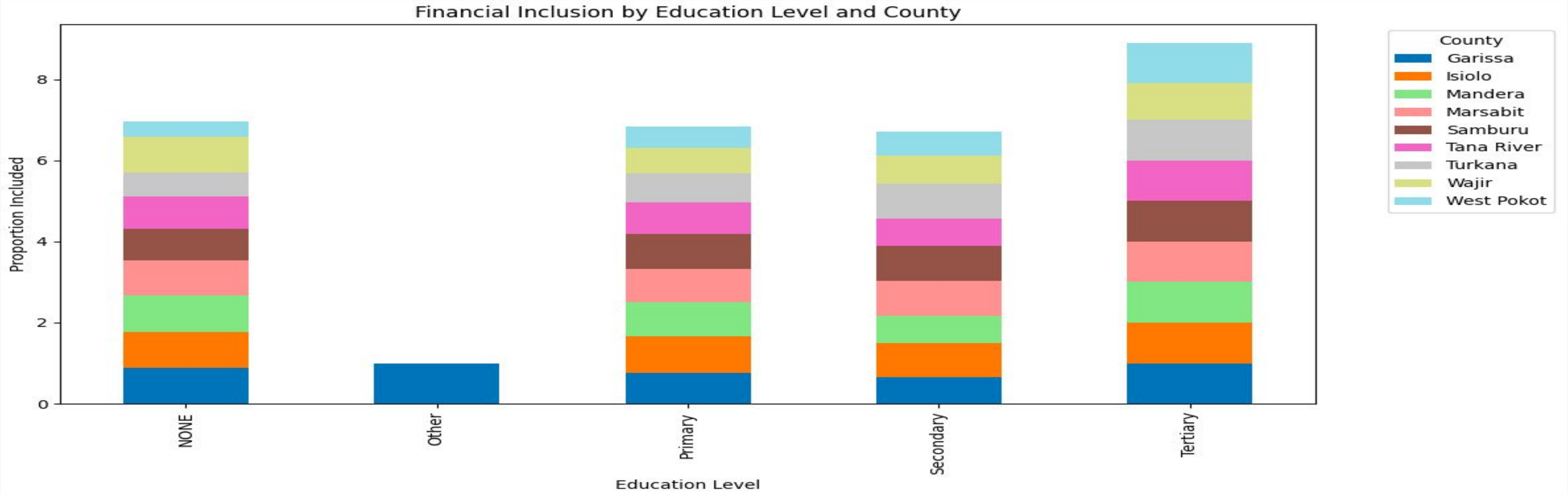




Across all the counties shown, way more people are financially included than those who are not (Wajir (345 people) and Mandera (340 people) have the highest number of included people overall. West Pokot (173 people) and Turkana (133 people) have the highest count of people who are not included. Counties like Isiolo and Samburu show the best results in terms of keeping non-inclusion low, with only 34 and 48 people, respectively, in that category. This means nearly everyone in those two counties is financially included.

Males exhibit higher formal financial inclusion than females

Socioeconomic Drivers for Financial Inclusion

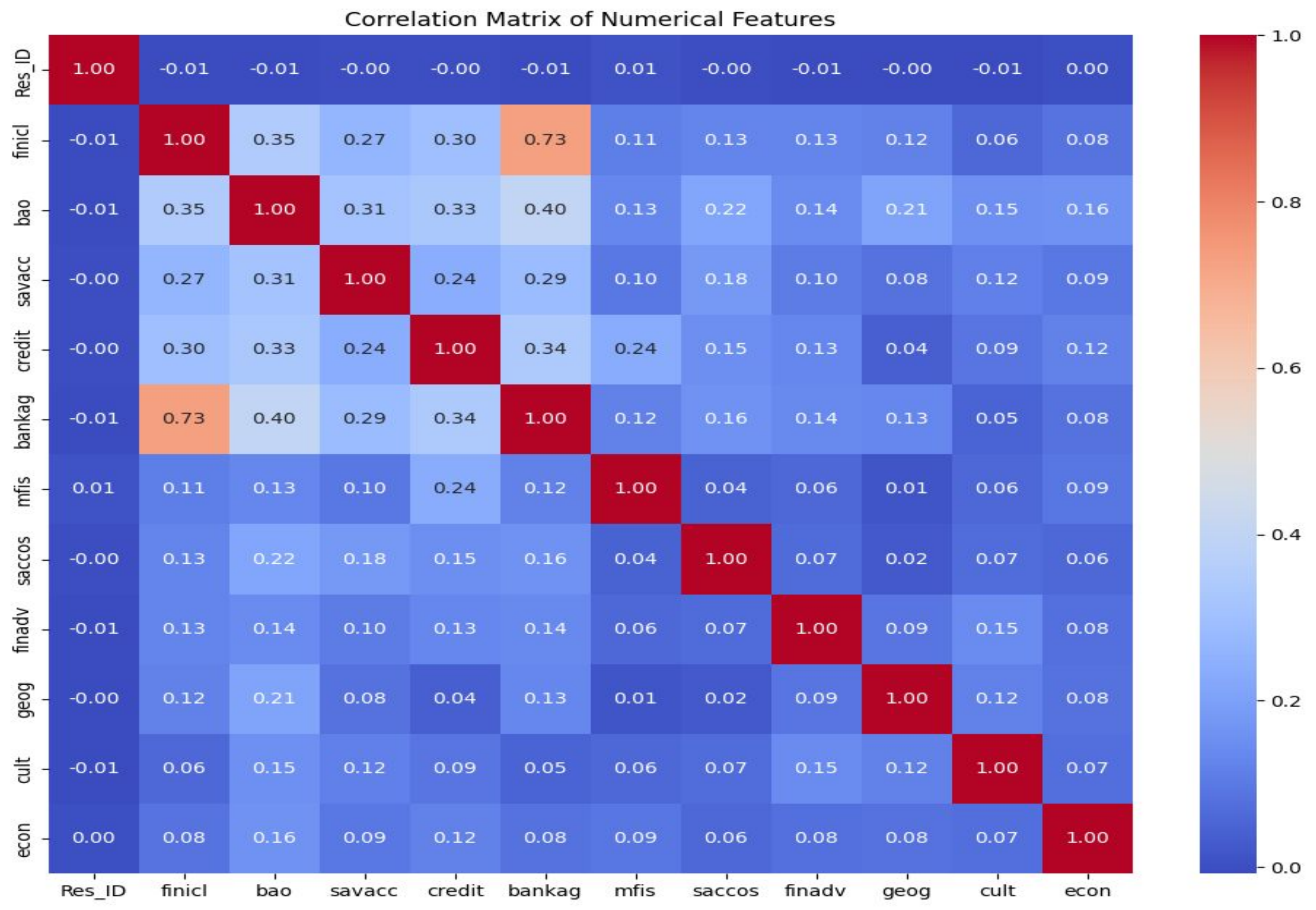


Financial inclusion increases with education level, with the highest inclusion observed among individuals with tertiary (college or university) education and the lowest among those categorized under "Other" education, which appears predominantly in Garissa County, suggesting it is either unique or underreported elsewhere.

Interestingly, individuals with no formal education also show a notably high inclusion rate, surpassing those with primary or secondary education, indicating that localized solutions such as mobile money platforms are effectively reaching less-educated populations.

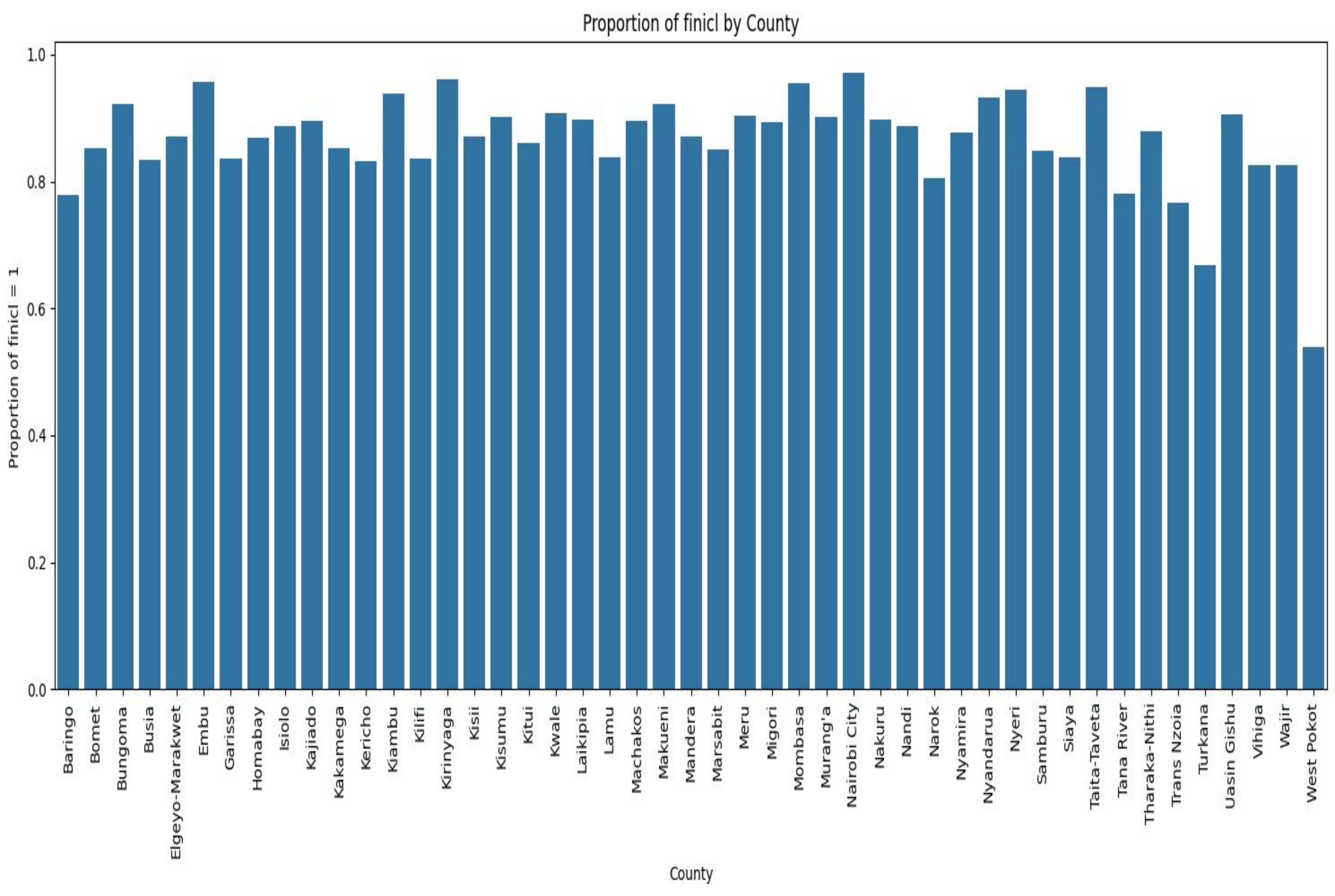
Across nearly all education levels, West Pokot, Wajir, and Turkana consistently emerge as the top counties in financial inclusion, reflecting broad access across education groups, while Isiolo and Marsabit maintain a steady, moderate level of inclusion throughout all education tiers.

Correlation matrix between financial and socioeconomic factors



- Financial inclusion indicator (finicl) and access to bank agents (bankag) show a high correlation (0.73), suggesting that increased financial inclusion is strongly associated with greater access to banking agents.
- Bank account ownership((bao) and Financial inclusion indicator (finicl) also correlate positively (0.35), indicating a link between banking outreach and financial inclusion.
- Financial inclusion indicator has a weak relationship with factors like geography and culture
  This means that financial inclusion is not highly dependent on these factors.

Proportion of Financial Inclusion By Counties in Kenya



- Proportion of the population in each county with access to formal financial services and products
- Significant disparities in financial inclusion across Kenya's counties indicating that financial services are not equally accessible to all Kenyans.
- Kiambu, Nairobi City, Embu and Kirinyaga show an average of 94% rate of inclusion.
- Counties like West Pokot and Turkana have the lowest inclusion rates of under 60%
- Counties with high inclusion rates are more urbanized or economically developed

Concept	How We Applied It
Analysis Parameters	Core questions & configuration settings. E.g., *Which factors predict formal financial inclusion in the informal sector?*
Cohorts	Entity subsets: e.g., *mobile-only traders*, *multi-channel users*, *female-led kiosks*.
Vohorts	Variable subsets: e.g., *mobile adoption features*, *business turnover & cash-flow metrics*.
Query Results	Clean derivative tables from structured SQL queries on the combined survey + FinAccess data.
Summaries	County-level stats for each cohort, such as mean mobile-money usage or average distance to nearest agent.
Comparisons	Cross-county, gender, and age contrasts of financial inclusion rates.
Similarities	Nearest-neighbor analysis to identify counties with similar inclusion profiles.
Correlations	Variable relationships—e.g., education vs. inclusion probability.
Descriptive Models	EDA visuals and clustering to describe current patterns.
Projection Models	Unsupervised clustering projecting new or emerging cohorts.
Predictive Models	**Probit regression** estimating probability of formal inclusion based on key predictors.
Systems Models	Linking cohorts & vohorts with external policy indicators (agent density, mobile coverage).

Key Findings

Access Gap: Mean formal inclusion across study counties ~**58%**, far below national 84.8%.

Top Predictors:

- **Mobile phone ownership** (strong positive, $p<0.01$)
- **Education level** (positive, $p<0.05$)
- **Distance to financial access point** (negative, $p<0.05$)

Barriers: ID registration issues, sparse agent networks, high cost of digital credit.

Qualitative Insight: Traders perceive mobile loans as costly; prefer rotating savings groups.

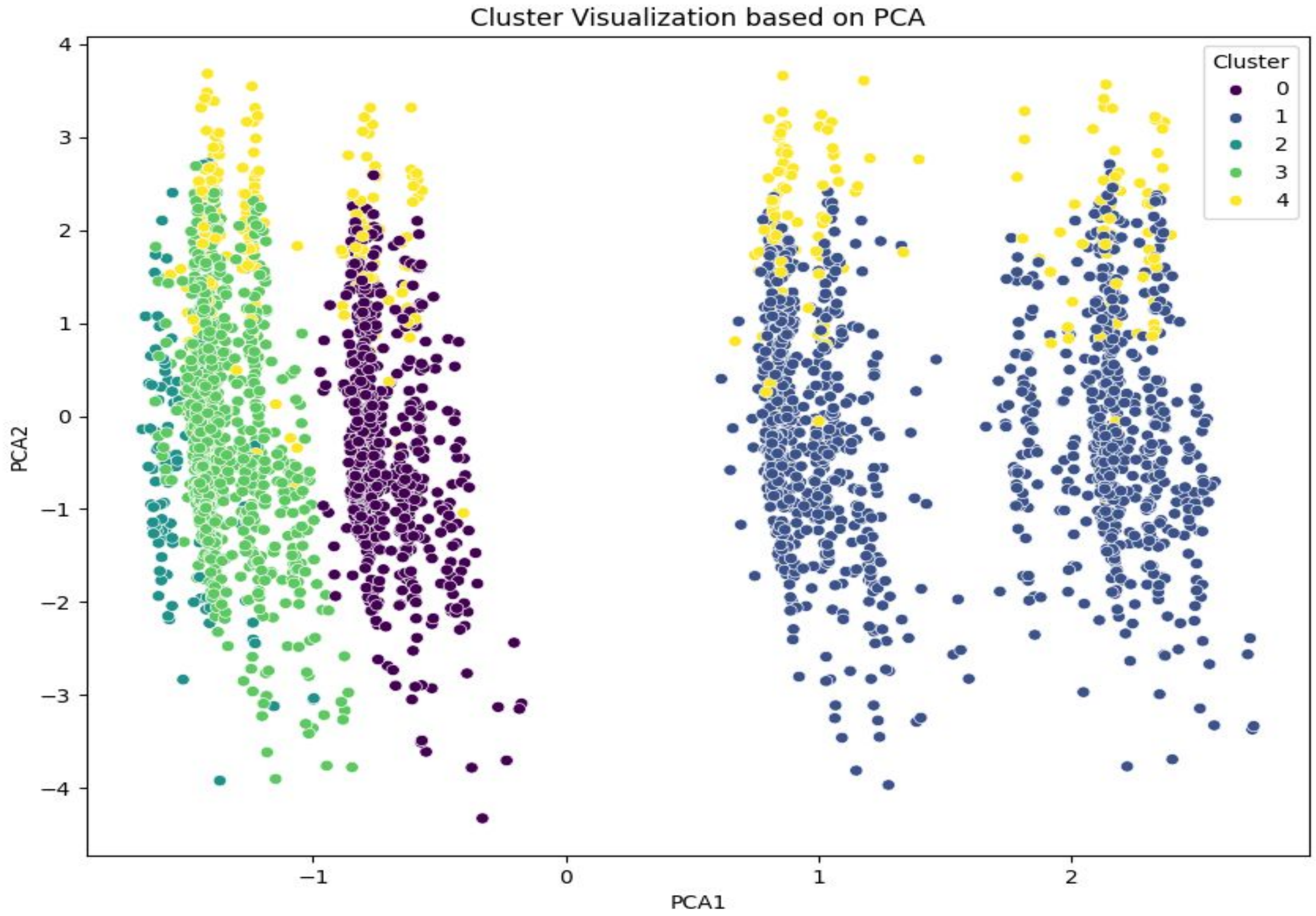
Financial Behavior Segmentation Using K-Means Clustering

Objective: To segment the population based on financial behaviors and access patterns.

Methodology: K-Means clustering on scaled data, with PCA for visualization.

Identified Clusters: We identified 5 distinct clusters with the following characteristics.

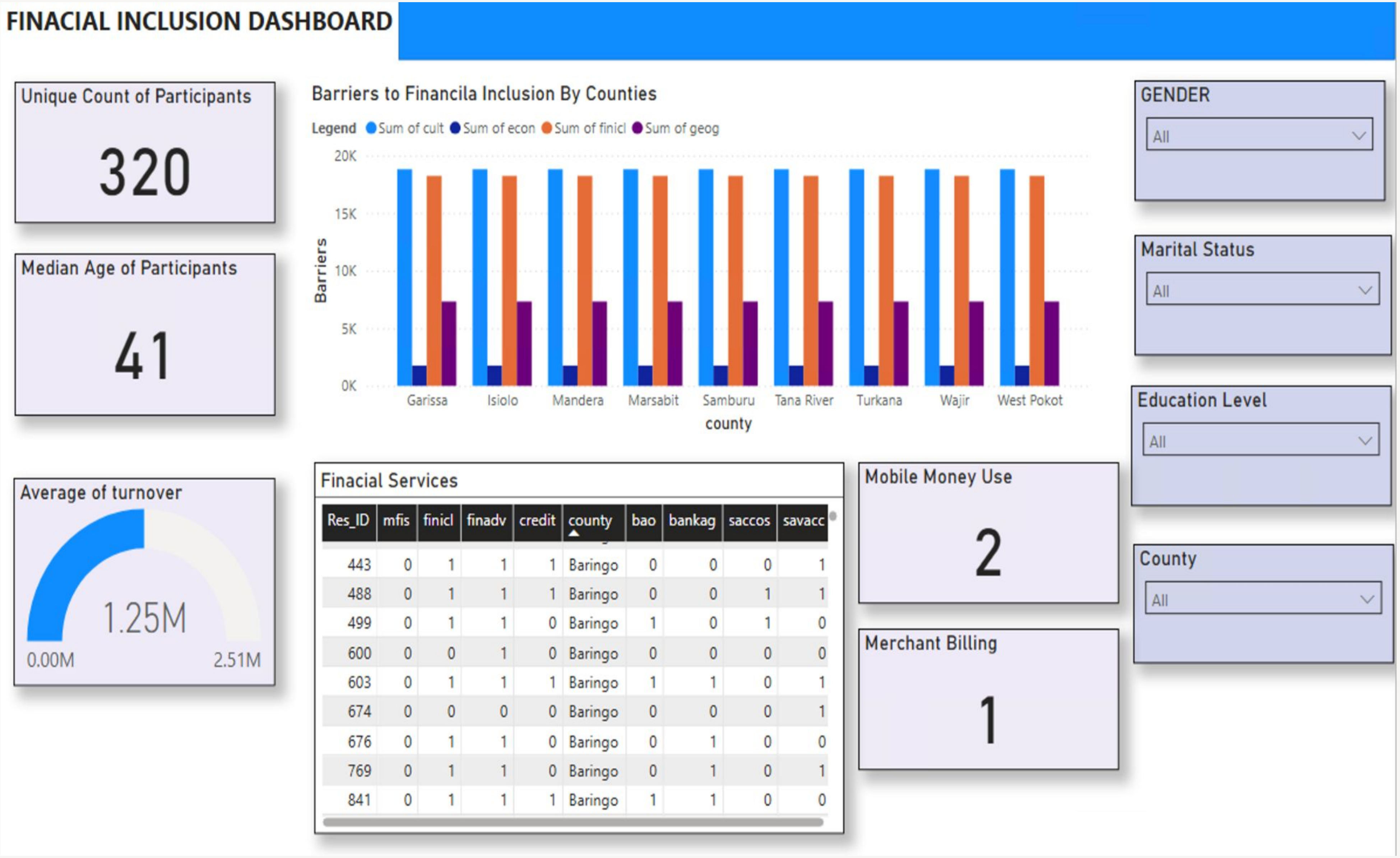
Cluster	Description
Cluster 0	Younger, ID Card holders, high Bank Transfer users.
Cluster 1	High Mobile Money users, some Bank/SACCO/MFI Paybill usage.
Cluster 2	High Bank Cheque and Bank Transfer users (likely business-oriented).
Cluster 3	Primarily Bank/SACCO/MFI Paybill users.
Cluster 4	Youngest demographic, 100% ID Card holders, nascent digital financial service users.



The PCA plot visually demonstrates the separation of these distinct groups.

DASHBOARD

Final Data Product



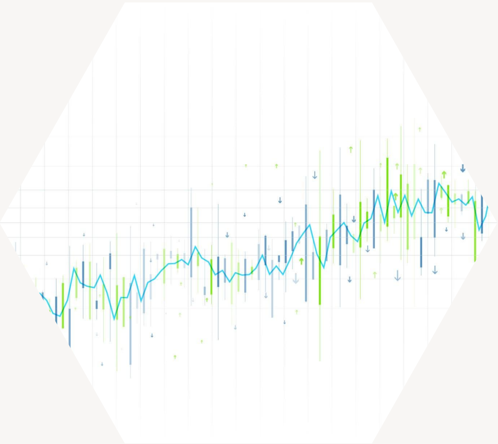
The data points are thus referred to in the visual board, we have analysed various facets of the Fin Survey data.

Policy & Fintech Recommendations

We recommend the following;

- Expand ID Registration: mobile units to remote areas.
- Strengthen Agent Networks: incentivize coverage in sparsely populated zones.
- Tailored Credit Products: flexible repayment to match seasonal incomes.
- Financial Literacy Programs: community-led for cultural relevance.

Next Steps



Extend to time-series analysis using FinAccess 2021–2024 for trend forecasting.

Integrate telco transaction logs for richer behavioral insights.

Collaborate with SACCOs/fintechs to pilot targeted interventions.

Thank You / Q&A

Link or QR code to live dashboard demo

Link or QR code to your Power BI dashboard.

Quick walk-through:

select a county → adjust predictors → watch probability curves update.